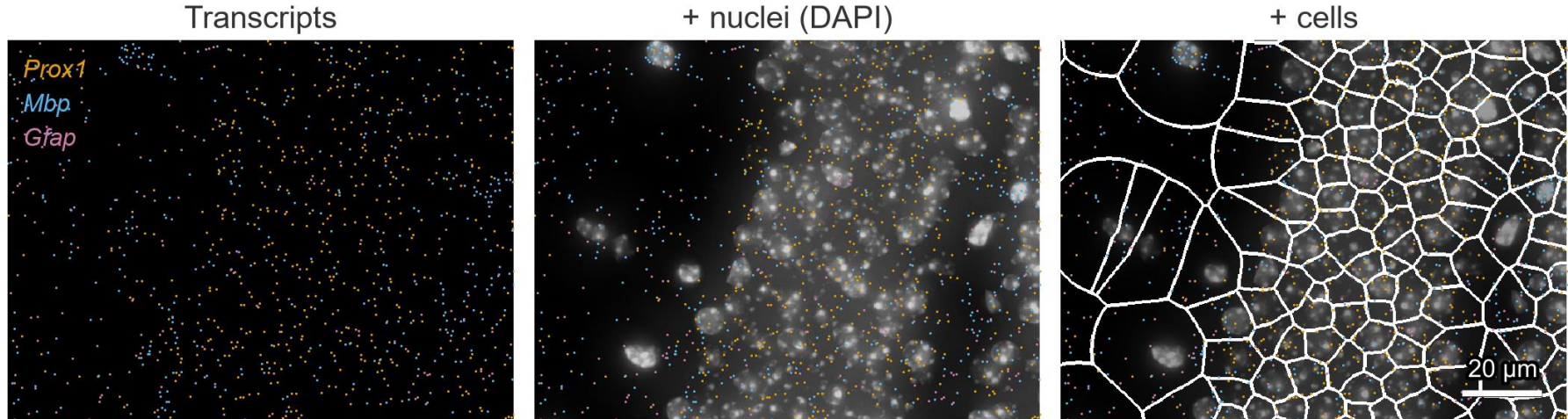


Cell segmentation

Christophe Avenel, NBIS
ELIXIR SCO Spatial Omics Data Analysis, Uppsala, 6 October 2026

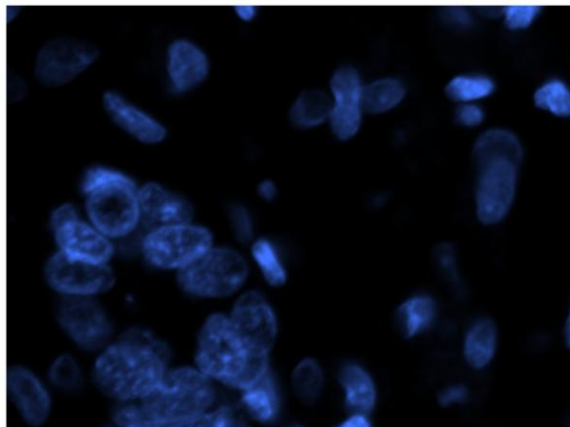
Why segment?



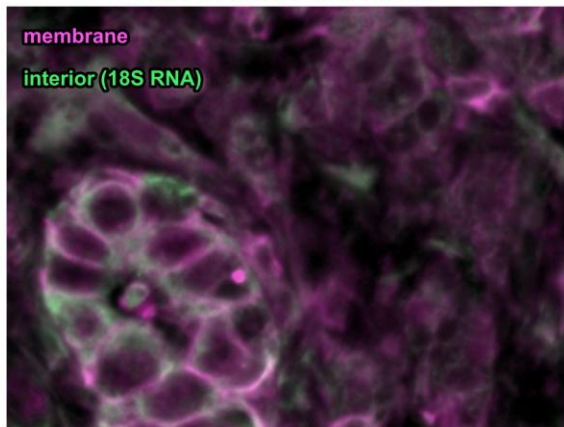
The instrument measures molecules. To get one row per cell, we need cell outlines.

What can we use to find cells?

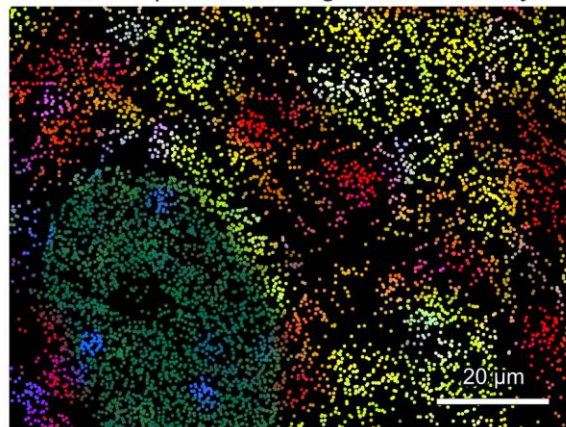
Nuclei: DAPI



Whole cells: cell stains



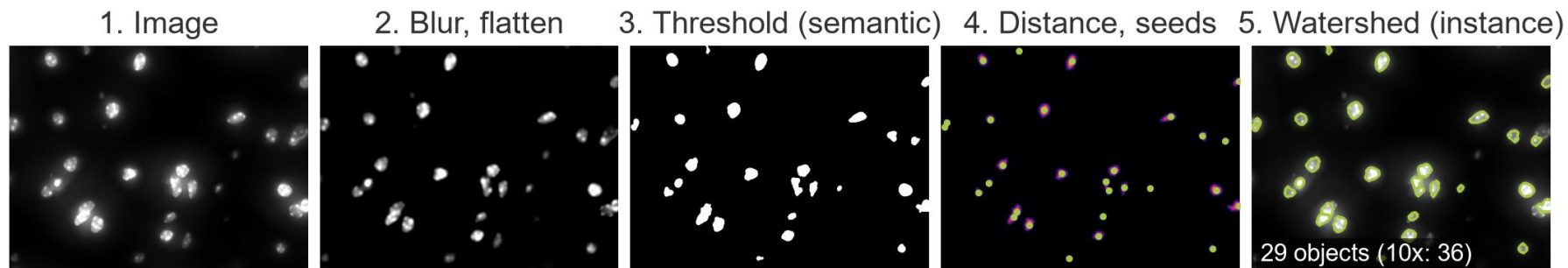
Transcripts: colour = gene mix nearby



Best signal: membrane stain > interior stain > nucleus + expansion. The transcripts can help too.

Practical 1: an older Xenium run with DAPI only, so we segment nuclei.

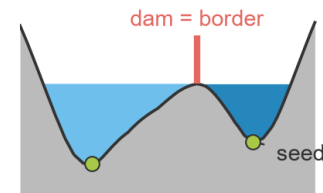
Threshold and watershed



Semantic: a class per pixel (step 3). **Instance:** one number per nucleus (step 5).

Works well when nuclei are bright and apart.

Fast, no training, every step can be checked.



Watershed in dense tissue

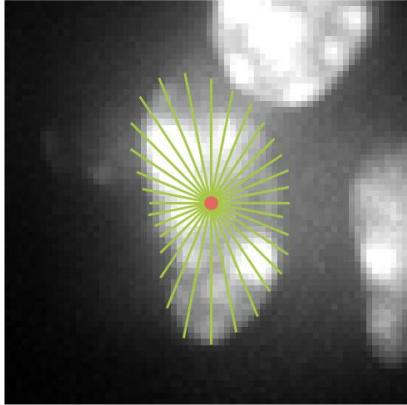


Fails when nuclei touch: it finds bright spots inside nuclei, not whole nuclei.

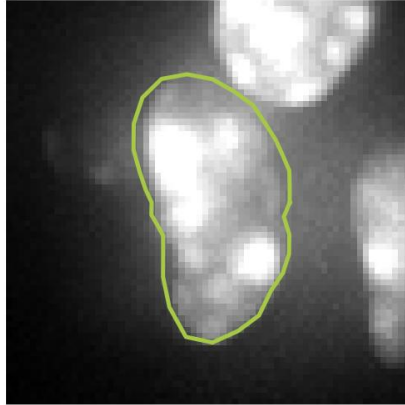
Settings that fix one region break another.

StarDist

32 distances from the centre



Polygon



- For each pixel: probability of being inside, and the distance to the border in 32 directions
- Star-shaped polygons: good for round nuclei, not for irregular cells
- Fast on a CPU.
Pretrained models for fluorescence and H&E
- Used in Space Ranger 4 (Visium HD) and bin2cell

Cellpose

1

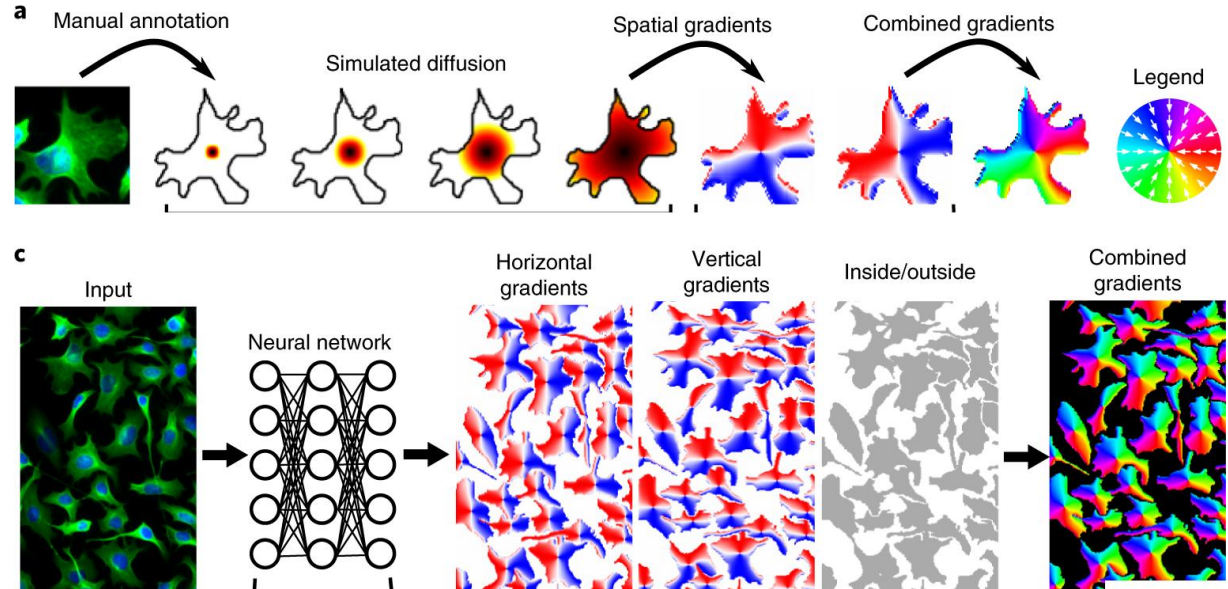
From each training mask, compute flows that point to the cell centre

2

A U-Net learns to predict these flows from the image

3

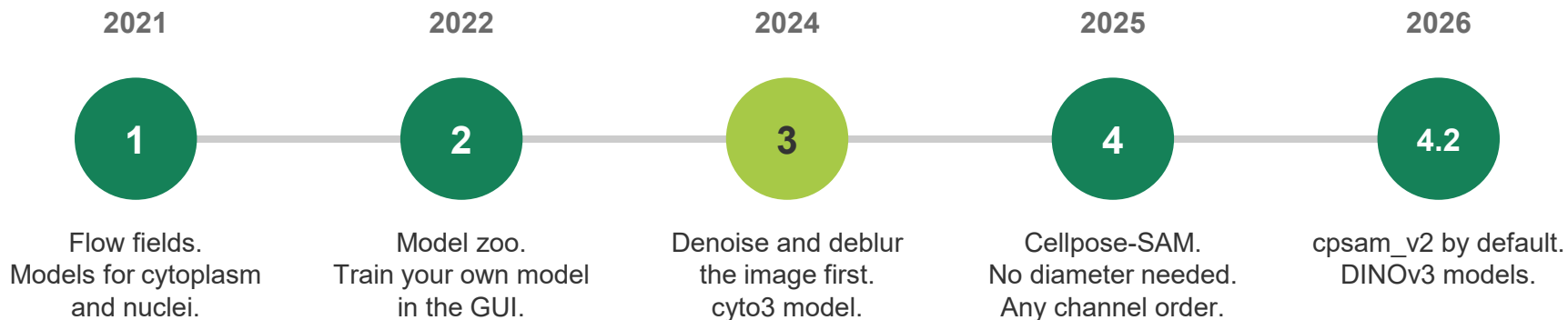
Pixels follow the flows. Pixels that meet form one cell



Nuclei and whole cells: give it a cell stain and a nucleus channel, or the nucleus channel alone.



Cellpose versions

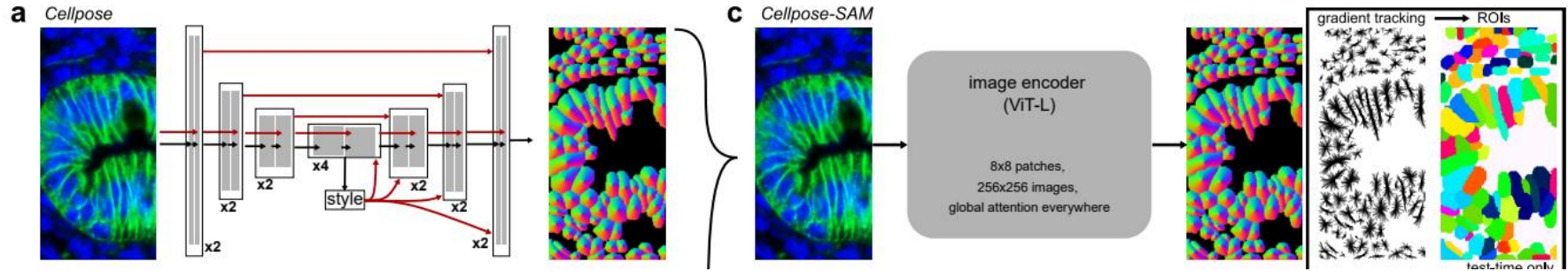


Practical 1: 3.1.0

Watch out: Cellpose 4 ignores `model_type="nuclei"` and runs Cellpose-SAM.

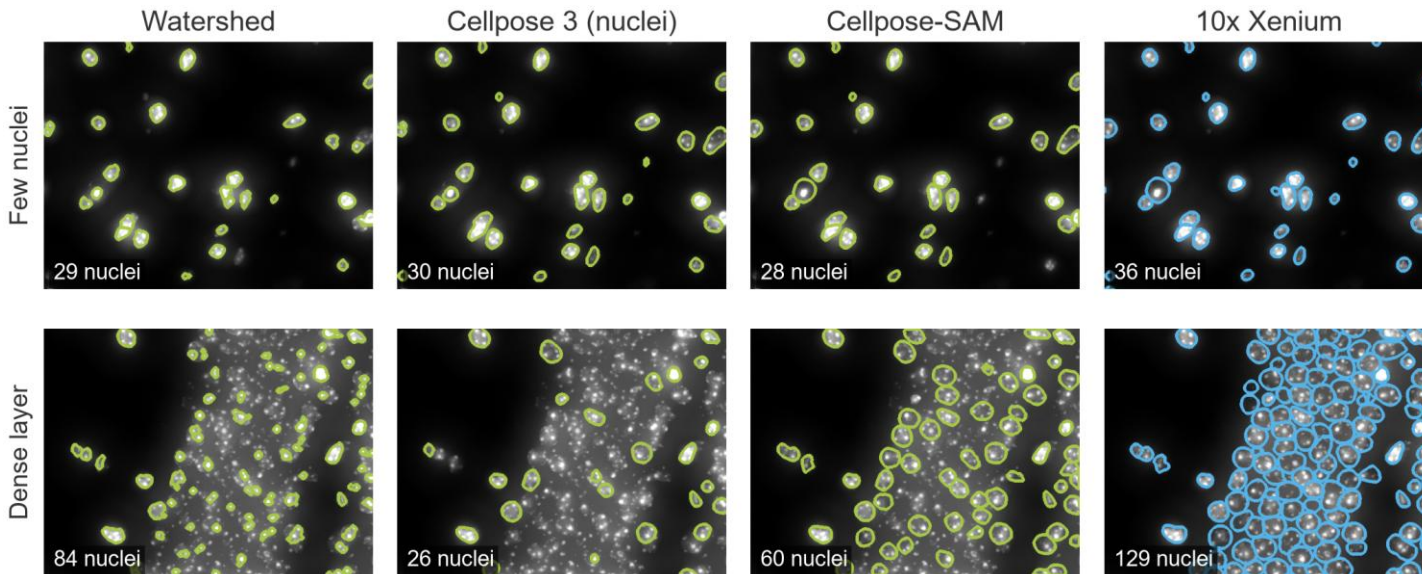


Cellpose-SAM



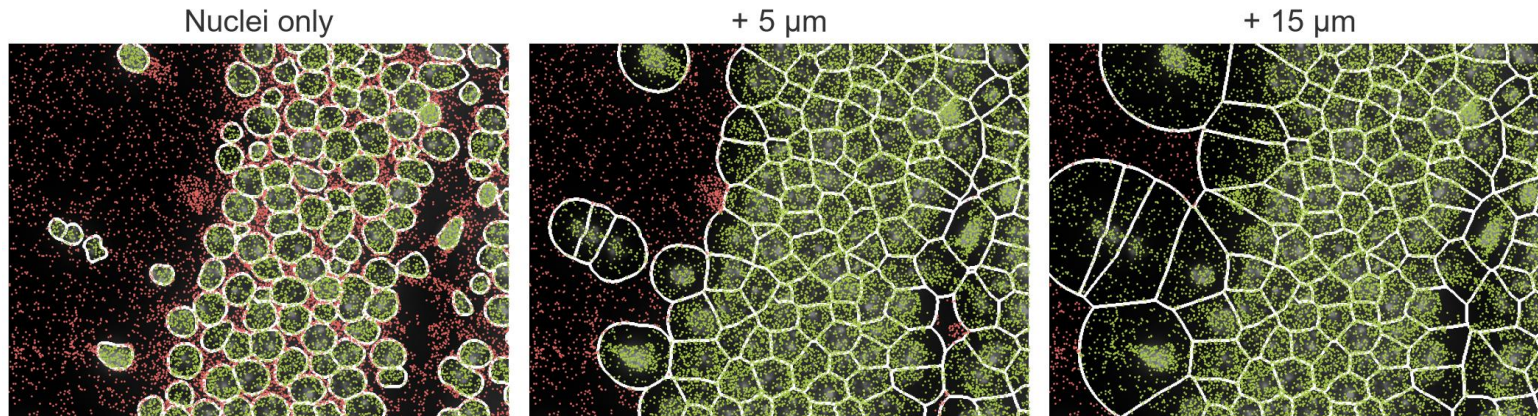
- Same flows as Cellpose. The U-Net is replaced by the image encoder of Segment Anything
- Up to 3 channels, any order (e.g. membrane, interior, DAPI). No diameter needed
- About 50 times more parameters than Cellpose 3. **Use a GPU.**

Nuclei: same image, four methods



The methods agree on sparse nuclei, but not in the dense layer.

From nuclei to cells: expansion



Pipeline

Xenium, XOA before 2.0 (our data)

Xenium, XOA 2.0 and later

Visium HD, Space Ranger 4

Default expansion

15 μm

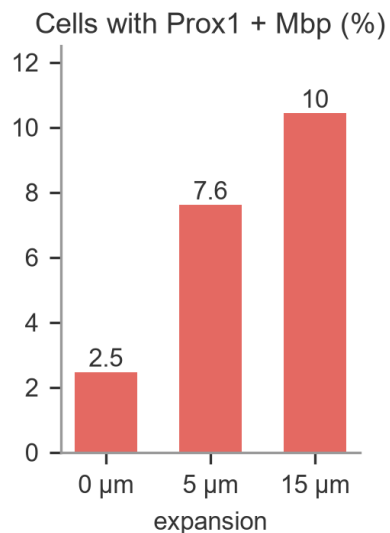
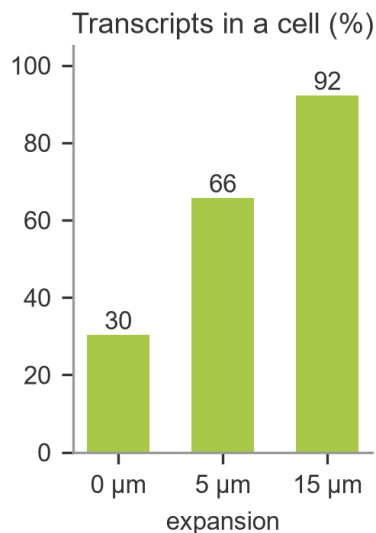
5 μm

8 μm

Expansion assumes:

the nucleus is in the centre,
cells are round,
all cells have the same size.

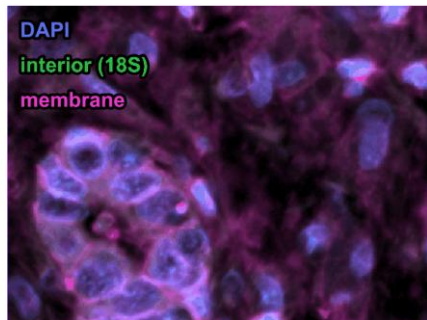
Bigger cells, more mixing



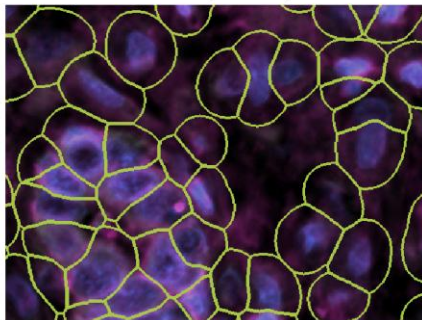
- More expansion: more transcripts in cells, but also transcripts from the neighbours
- *Prox1* (granule neurons) and *Mbp* (oligodendrocytes) should not be in the same cell
- This creates false co-expression: "spatial doublets"
- Correction after segmentation: cellAdmix, SPLIT, ResolVI

Cell stains or nucleus expansion?

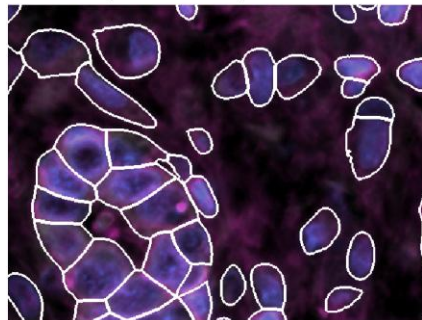
Stains



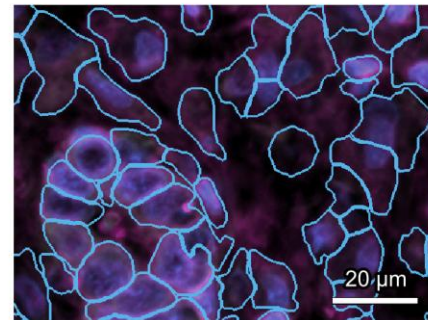
Nucleus + 5 μ m expansion



Cellpose-SAM, cell stains

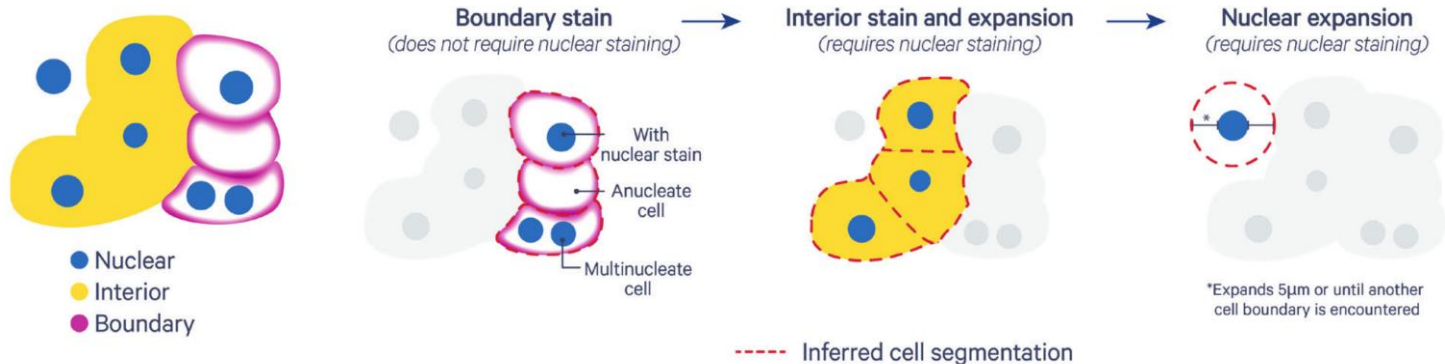


10x multimodal



Expansion makes round cells and fills the gaps. **Cell stains follow the real shapes.**

Xenium multimodal segmentation



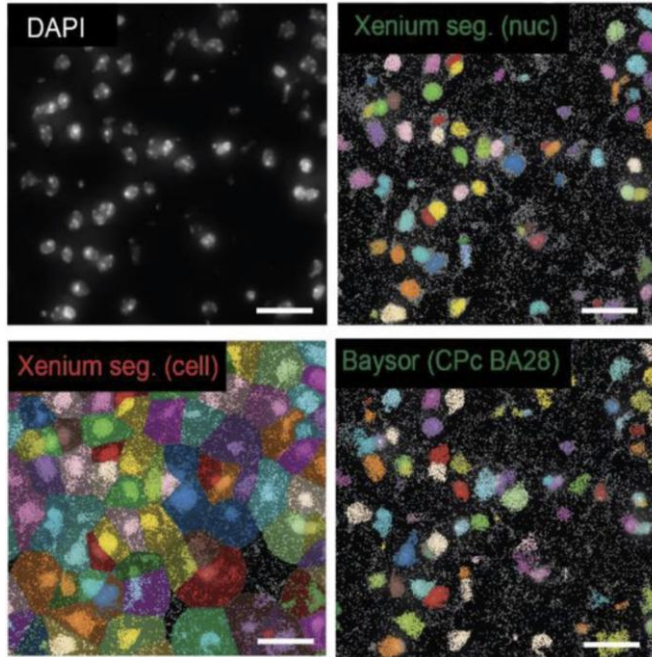
Mouse
liver



One method per cell, stored in `cells.parquet`.

Expansion splits the binucleate cells (arrows).
The boundary stain does not.

Use the transcripts too



- Transcripts from one cell have a similar gene mix
- **Baysor**: Bayesian model of the molecules, can start from nuclei
- **Proseg**: probabilistic model of cells; best in the Bilous et al. 2026 Xenium benchmark
- **Segger**, **BIDCell**, **ComSeg**: graph and deep learning methods
- Sections are 5-10 μm thick: cells can overlap, so even a perfect 2D outline can mix two cells

What your platform does

Platform	Cell stains	Default segmentation	Run again / import your own
Xenium	membrane (ATP1A1, CD45, E-cadherin), interior (18S)	per cell: boundary, interior, or 5 μ m expansion	Xenium Ranger
MERSCOPE	cell boundary kit (3 proteins), PolyT	Cellpose on DAPI and cell stains	VPT (open source)
CosMx	membrane (CD298/B2M), PanCK, CD45	Cellpose-based, membrane stain	AtoMx
Visium HD	H&E	StarDist + 8 μ m expansion	Space Ranger 4, bin2cell, ENACT

Other tools

Tool	Segments	Input	CPU	Used in
Watershed	nuclei	1 channel	yes	scikit-image, QuPath
StarDist	nuclei	DAPI, H&E, 3D	yes	QuPath, napari, sopa, Space Ranger
Cellpose 3	nuclei or cells	1 or 2 channels	yes	napari, sopa, Vizgen VPT
Cellpose-SAM	nuclei or cells	up to 3 channels	slow	napari, sopa
InstanSeg	nuclei and cells	any channels, H&E	yes, fast	QuPath, sopa
Mesmer	nuclei and cells	nucleus + membrane	yes	DeepCell
CellSAM	cells	IF, brightfield	GPU	Python, web app
micro-SAM	interactive	light and EM	GPU	napari

No single tool wins on all data.

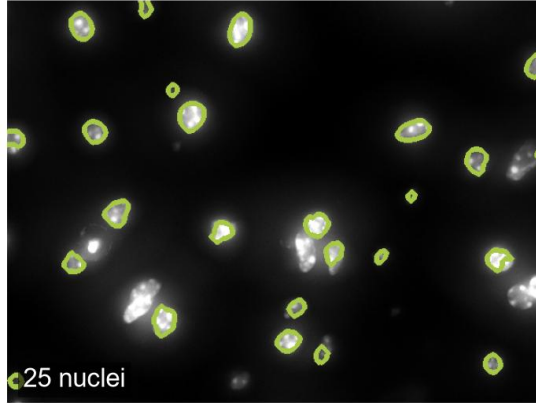
Running it on many samples

Pipeline	Data	What it runs	Run with
sopa	any imaging platform, Visium HD	segmentation in patches (Cellpose, Baysor, Proseg...), counts, annotation	Python, Snakemake, nf-core/sopa
nf-core/spatialaxe	Xenium	re-segmentation (Cellpose, Baysor, Proseg, Segger), Xenium Ranger, QC	Nextflow
nf-core/spatialvi (in development)	Visium, Visium HD	Space Ranger, QC, clustering, spatially variable genes, report	Nextflow

All three write SpatialData. The practicals show each step by hand; for many samples, use a pipeline.

Cellpose 3: the parameters you will change

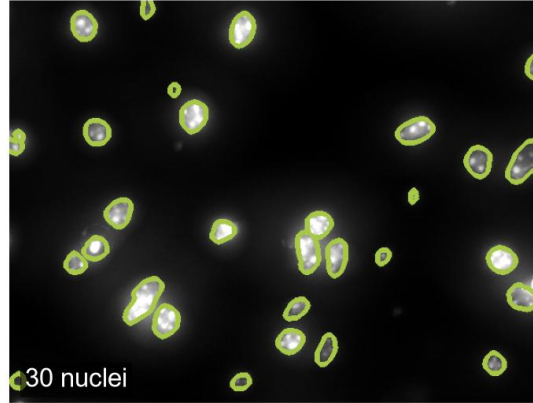
diameter 2 μm (10 px)



diameter

in pixels = size in μm / pixel size
0.2125 μm at level 0, 0.425 μm at level 1

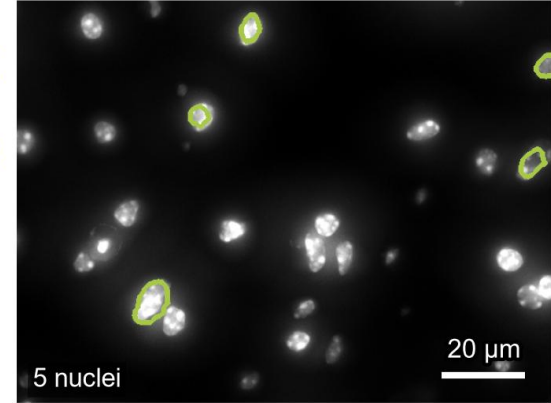
diameter 6 μm (30 px)



flow_threshold

default 0.4. Raise it if masks are missing,
lower it if shapes look wrong.

diameter 17 μm (80 px)



cellprob_threshold

default 0. Lower it to find dim nuclei,
raise it to remove background.

How good is my segmentation?

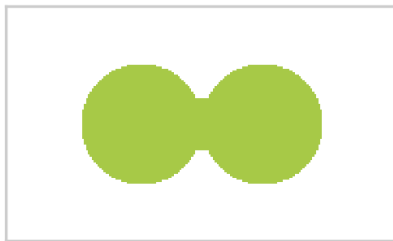
Reference: 2 nuclei



0.99

Dice (pixels)

Prediction: 1 object



0

F1, objects matched at IoU 0.5

- Pixel scores (Jaccard, Dice) do not see that two nuclei became one
- Object scores pair each object with one reference object (IoU of at least 0.5)
- Practical 1 compares with the 10x segmentation, which is a model output too



Summary

- 1 Segmentation turns molecules into cells. Every later step depends on it.
- 2 Use the best signal you have:
membrane > interior > nucleus + expansion.
- 3 Learned models beat thresholds, but dense tissue is still hard.
- 4 Expansion is a trade-off: more transcripts, more mixing.
- 5 There is no ground truth. Compare methods and look at the images.

Next

Practical 1: Cellpose 3
on the Xenium mouse brain

Tomorrow

Naveed Ishaque:
what if we skip the cell?

Hint

Save your Cellpose labels
(notebook section 6) and load
them in TissUMaps:
go.nbis.se/tissuumaps